What the equations of loop quantum gravity give us is the probability associated with a given possible *boundary* of the box – the probability that the balls will come out of the box in one particular configuration or another, if they have entered it in another.

How is this probability computed? Recall Feynman's sum over paths, which I described when speaking about quantum mechanics. Probabilities, in quantum gravity, can be calculated in the same way. By considering all the possible 'trajectories' that have the same boundary. Since we are including the dynamics of spacetime, this means considering *all possible spacetimes* which have the same boundary as the box.

Quantum mechanics assumes that between the initial boundary, where the two balls enter, and the final boundary where they exit, there is no definite spacetime nor definite trajectory of the balls. There is a quantum 'cloud' in which all the possible spacetimes and all possible trajectories exist together. The probability of seeing the balls going out in one way or another can be computed by summing over all possible spacetimes.

Spinfoam

If quantum *space* has the structure of a spin network, what structure will *spacetime* have? What will one of the spacetimes previously alluded to in the calculation be like?

It must be a 'history' of a spin network. Imagine that you take the graph of the spin network and move it: every node in the web draws a line, like the balls in figure 7.1, and every line of the graph, moving, draws a surface (for example, a moving segment draws a rectangle). But there is more: a node can open up into two or more nodes, just as a particle can split into two or more particles. Conversely, two or more nodes can combine into a single one. In this way, a graph which evolves draws an image like the one in figure 7.2.